

SIMATS ENGINEERING (SAVEETHA SCHOOL OF ENGINEERING)

Department of Artificial Intelligence and Data Science

ASSESSMENT TOOL 2 – Industry Problem Solving Task

Course Code:	DSA03	Course Title:	Natural Language Processing
CO Assessed:	CO4 – Precision	Assessment:	Assessment Tool 2 – Industry Problem Solving Task
Weightage:	40% of CIA		
CO5 Statement:	Formulate context-free grammars, feature structures, probabilistic parsing techniques and to construct parsing frameworks. (BL-4)		
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Section / Batch:	BE CSE AI / 2024	Date:	13-08-2026

Code of Conduct

/ NARAPPAGARI SRAVYA(192472192) (Name / Reg No) certify that this submission is my original work and that I have adhered to the guidelines specified for this assessment. I understand that any violation of academic integrity rules will result in disciplinary action.

Signature of the Student: SRAVYA

Instructions

- Read each industry scenario carefully before answering.
- Analyze the given problem from a semantic analysis perspective.
- Provide structured and logical answers with proper justification.
- Support your analysis using examples from the given scenario wherever applicable.
- Use suitable diagrams, semantic representations, logical predicates, workflows, architectures, or tables whenever necessary.
- Clearly state assumptions if additional information is required.
- Duration: 30 minutes.

Section / Topic	Focus	Question No.
Representing Meaning & Meaning Structure of Language	Analyze semantic interpretation of user queries, identify ambiguity in meaning representation, evaluate semantic processing limitations, and improve understanding in banking chatbot applications.	Q1

Syntax-Driven Semantic Analysis	Analyze multiple interpretations of spoken commands, evaluate parsing-related ambiguity, and assess semantic extraction in real-time voice assistant systems.	Q2
First-Order Predicate Calculus, Representing Linguistically Relevant Concepts, Syntax-Driven Semantic Analysis	Design a semantic processing architecture for healthcare reports, model relationships among entities and actions, represent linguistic concepts, and improve semantic extraction accuracy.	Q3

Annexure A – Industry Problem Solving Task

1. A company is developing an AI-powered customer support chatbot for banking services. The chatbot uses Context-Free Grammar (CFG) for parsing user queries. However, it faces issues such as:

Incorrect interpretation of ambiguous sentences

Failure to handle subject–verb agreement Inefficient parsing for long queries Example user query:

“Show me the transactions with the card from last month”

- Analyze the ambiguity present in the given sentence using CFG
- Evaluate the limitations of CFG in handling real-world conversational queries
- Evaluate and propose an improved solution using PCFG, feature structures, or Earley parsing
- Justify how your proposed method improves accuracy and efficiency in an industry setting

2. A voice assistant system (like Alexa/Siri-type applications) processes spoken commands using parsing techniques. The system currently uses top-down parsing, but faces:

Backtracking issues

Poor handling of incomplete or ambiguous input

Difficulty in real-time response Example command:

“Book a flight to Delhi with a window seat”

- Analyze how the given sentence can lead to multiple parse structures
- Analyze the limitations of top-down parsing in this real-time application
- Analyze how Earley parsing can improve handling of ambiguity and partial inputs
- Compare and analyze the performance of top-down vs Earley parsing for such industry use cases

3. A healthcare company is developing an AI system to analyze patient medical reports and automatically extract critical information such as diagnosis, prescribed treatments, and follow-up actions.

The system must handle:

Complex and lengthy medical sentences

Ambiguity in sentence structure

Accurate subject-verb agreement

Relationships between medical actions and entities (sub-categorization) Real-time processing for hospital use Example input:

"The doctor who reviewed the patient last week recommends starting medication and scheduling a follow-up visit in Chennai."

The current system, based only on basic Context-Free Grammar (CFG), fails to correctly interpret relationships and produces inconsistent outputs.

a) Design a complete NLP architecture integrating:

CFG for syntactic structure

Probabilistic CFG (PCFG) for ambiguity resolution

Feature Structures for agreement handling

Efficient parsing techniques

b) Develop a step-by-step workflow showing how the system processes the given medical sentence from input to structured output (diagnosis, action, location). c) Create a method to:

Incorporate sub-categorization frames for medical verbs

Handle real-time data processing

Improve accuracy and scalability in a hospital environment

Rubrics for Calculation

Criteria		Weight age	Level 4 – Exemplary		Level 3 – Proficient		Level 2 – Developing		Level 1 – Beginning	
Understanding of Industry Problem		20%	Comprehensive understanding of real-world problem and constraints		Good understanding with minor gaps in requirements		Basic understanding; some requirements unclear		Poor understanding ; misinterprets problem context	
Application of Engineering Concepts		25%	Applies advanced and relevant concepts effectively to solve the problem		Applies appropriate concepts with minor errors		Limited application of concepts		Incorrect or no application of concepts	
Solution Design		25%	Innovative, practical, and feasible solution aligned with industry standards		Logical and feasible solution with minor gaps		Basic solution with limited feasibility		Unclear or impractical solution	
Use of Tools		15%	Effective use of modern tools, technologies, and real data		Appropriate use of tools with correct implementation		Limited or inconsistent use of tools		Minimal or incorrect use of tools	
Analysis		10%	Thorough analysis with validated results and performance evaluation		Good analysis with reasonable validation		Basic analysis with limited validation		Poor or no analysis	
Professionalism		5%	Highly professional, well-documented, and clearly presented work		Good documentation and presentation		Basic documentation with some gaps		Poor documentation and presentation	
Q. No. / Criteria Level	Understanding of Industry Problem	Application of Engineering Concepts	Solution Design	Use of Tools	Analysis	Professionalism	Sub. Total	Total %		
1										
2										
3										
Faculty In-charge			Course Coordinator			Program Director				
Signature: _____			Signature: _____			Signature: _____				
Date: _____			Date: _____			Date: _____				

1. AI-Powered Banking Customer Support Chatbot

Given sentence:

“Show me the transactions with the card from last month.”

a) Ambiguity using CFG

The sentence has structural ambiguity because the phrase “**with the card**” can have different attachments.

Interpretation 1:

- The user wants transactions **made using the card**.
- Structure: transactions → with the card

Interpretation 2:

- “With the card” could be interpreted as information associated with the transactions/card rather than directly modifying the transaction action.

A simplified CFG can represent:

$S \rightarrow VP$

$VP \rightarrow V NP$

$NP \rightarrow Det N PP$

$PP \rightarrow P NP$

Possible parsing:

Show [me [the transactions [with the card] [from last month]]]

The main ambiguity is deciding which noun or phrase “**with the card**” and “**from last month**” modify.

b) Limitations of CFG in real-world conversations

Basic CFG has several limitations:

1. **Cannot easily represent probabilities** between multiple parses.
2. **Poor handling of ambiguity** in natural language.
3. **Does not naturally represent subject–verb agreement**.
4. **Does not capture conversational context**.
5. Long sentences can produce many possible parse trees.
6. It requires many grammar rules for different sentence patterns.
7. It struggles with incomplete or informal user queries.

Therefore, a basic CFG alone is insufficient for a production banking chatbot.

c) Improved solution

A combination of **PCFG + Feature Structures + Earley Parsing** can be used.

- **PCFG:** Assigns probabilities to different parse trees and selects the most likely interpretation.
- **Feature Structures:** Represent grammatical features such as number, tense, person, and agreement.
- **Earley Parsing:** Efficiently handles ambiguous and incomplete sentences and works well with general CFGs.

For example, the system can assign a higher probability to:

“transactions made using the card”

than an unlikely alternative interpretation.

d) Industry benefits

The proposed approach improves:

- **Accuracy:** PCFG selects the most probable interpretation.
- **Grammar correctness:** Feature structures handle agreement.
- **Efficiency:** Earley parsing avoids unnecessary repeated parsing.
- **Robustness:** Better handling of long and incomplete queries.
- **Customer experience:** The chatbot can correctly retrieve the user's transactions.

2. Voice Assistant – Flight Booking

Given command:

“Book a flight to Delhi with a window seat.”

a) Multiple parse structures

The sentence can have different interpretations depending on how the phrase “**with a window seat**” is attached.

Parse 1 – Seat preference:

Book [a flight to Delhi] [with a window seat].

Meaning: The user wants a flight to Delhi and prefers a window seat.

Parse 2 – Flight description:

Book [a flight [to Delhi with a window seat]].

This treats the additional phrase as part of the description of the flight.

For a voice assistant, the first interpretation is normally the intended one because “**window seat**” represents a passenger preference.

b) Limitations of top-down parsing

Top-down parsing starts from the start symbol and attempts to generate the input.

Problems include:

- **Backtracking:** It may try several alternatives before finding the correct parse.
- **Repeated work:** The same partial structures may be analyzed multiple times.
- **Poor ambiguity handling:** Multiple interpretations can increase computation.
- **Incomplete input:** Spoken commands may stop unexpectedly.
- **Real-time constraints:** Excessive backtracking can increase response latency.

For voice assistants, even small delays can negatively affect user experience.

c) Earley parsing advantages

Earley parsing is suitable because it can efficiently handle **general CFGs, ambiguity, and partial input**.

It uses three major operations:

1. **Prediction** – predicts possible grammar rules.
2. **Scanning** – matches grammar symbols with input words.
3. **Completion** – completes partially recognized structures.

For:

“Book a flight to Delhi with a window seat”

Earley parsing can maintain multiple possible interpretations without completely restarting the parsing process.

It is also useful when speech recognition produces incomplete or uncertain input.

d) Top-down vs Earley Parsing

Feature	Top-Down Parsing	Earley Parsing
Ambiguity handling	Limited	Good
Backtracking	Often required	Reduced

Partial input	Poor	Good
General CFG support	Depends on parser	Yes
Long sentences	Can become inefficient	More suitable
Real-time applications	Less suitable for complex input	More suitable
Multiple interpretations	Difficult	Can maintain alternatives

Conclusion: For a voice assistant, **Earley parsing is generally preferable** when commands are ambiguous, incomplete, or structurally complex.

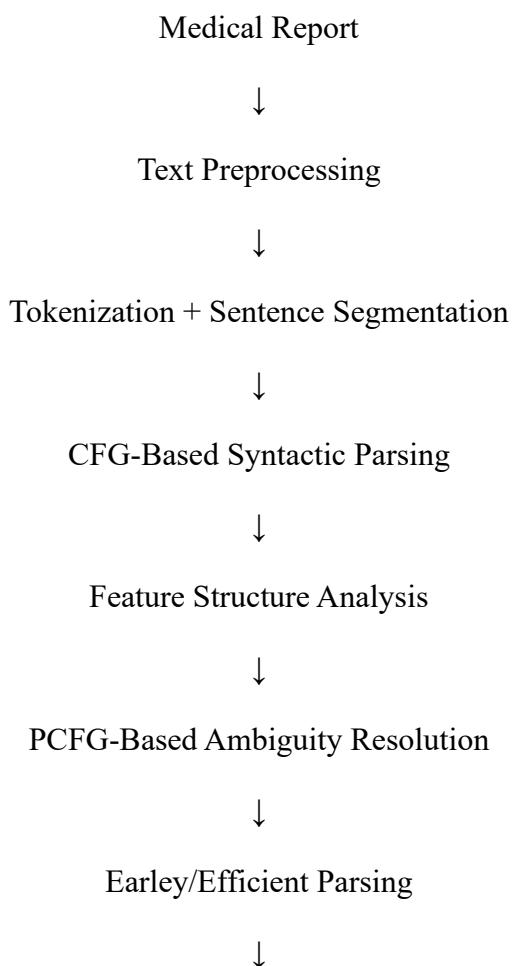
3. Healthcare Medical Report Analysis

Input:

“The doctor who reviewed the patient last week recommends starting medication and scheduling a follow-up visit in Chennai.”

a) Complete NLP Architecture

A suitable architecture is:



Medical Sub-Categorization



Named Entity Recognition



Semantic Role Analysis



Information Extraction



Structured Medical Output

1. CFG

CFG provides the basic syntactic structure.

Example:

$S \rightarrow NP VP$

$NP \rightarrow Det N$

$NP \rightarrow NP RelativeClause$

$VP \rightarrow V GerundPhrase$

It identifies relationships such as subject, verb, object, and clauses.

2. PCFG

PCFG assigns probabilities to competing parse structures.

For example, if a sentence has two possible interpretations, PCFG can select the interpretation with the highest probability based on training data.

3. Feature Structures

Feature structures can represent:

Person = 3

Number = Singular

Tense = Present

For:

“The doctor ... recommends”

both **doctor** and **recommends** are singular, so the agreement is valid.

4. Efficient Parsing

Earley parsing can be used because it supports general CFGs and handles ambiguous structures efficiently.

b) Step-by-Step Workflow

Step 1 – Input

The system receives:

“The doctor who reviewed the patient last week recommends starting medication and scheduling a follow-up visit in Chennai.”

Step 2 – Tokenization

The sentence is divided into tokens:

The | doctor | who | reviewed | the | patient | last | week |
recommends | starting | medication | and | scheduling |
a | follow-up | visit | in | Chennai

Step 3 – Syntactic Parsing

The system identifies:

- **The doctor** → Main subject
- **who reviewed the patient last week** → Relative clause
- **recommends** → Main verb
- **starting medication** → Recommended action
- **scheduling a follow-up visit** → Second recommended action
- **in Chennai** → Location

Step 4 – Feature Agreement

The system verifies:

Subject = doctor

Number = singular

Verb = recommends

Number = singular

Therefore:

doctor → **recommends** is grammatically consistent.

Step 5 – PCFG Disambiguation

If multiple parse structures are generated, PCFG assigns probabilities and selects the most likely structure.

Step 6 – Sub-Categorization

The system recognizes that:

recommend can take a complement describing an action:

recommend + starting medication

recommend + scheduling a follow-up visit

Step 7 – Named Entity Recognition

The system identifies:

- **Doctor** → Medical person
- **Patient** → Patient entity
- **Medication** → Treatment entity
- **Chennai** → Location

Step 8 – Structured Output

A simplified structured representation is:

Category	Extracted Information
Subject	Doctor
Diagnosis	Not explicitly stated
Action 1	Start medication
Action 2	Schedule follow-up visit
Location	Chennai
Time reference	Last week
Medical relation	Doctor recommends actions for patient

Important: The sentence does **not explicitly provide a diagnosis**, so the system should not invent one.

c) Sub-Categorization, Real-Time Processing and Scalability

1. Sub-Categorization Frames

Medical verbs should be associated with the types of arguments they normally accept.

Examples:

prescribe(Doctor, Medication, Patient)

recommend(Doctor, Action)

diagnose(Doctor, Disease, Patient)

schedule(Staff, Appointment)

report(Patient, Symptom)

For the given sentence:

recommend(Doctor, start(Medication))

recommend(Doctor, schedule(FollowUpVisit))

This prevents the system from incorrectly assigning medical entities to unrelated roles.

2. Real-Time Processing

For hospital environments:

- Use incremental/Earley parsing.
- Process reports as they arrive.
- Use optimized NLP models.
- Cache frequently used medical terms.
- Use parallel processing for multiple reports.
- Prioritize urgent medical information.
- Store structured results in the hospital information system.

3. Accuracy Improvements

Accuracy can be improved through:

- Medical-domain training data.
- PCFG-based ambiguity resolution.
- Feature structures for agreement.

- Medical ontologies.
- Named Entity Recognition.
- Semantic Role Labeling.
- Sub-categorization frames.
- Human validation for critical information.

4. Scalability

A scalable hospital system can use:

Multiple Hospital Reports



Processing Queue



Parallel NLP Workers



Parsing + Entity Extraction



Semantic Analysis



Central Medical Database



Doctor/Hospital Dashboard

This allows thousands of medical reports to be processed efficiently while maintaining consistent semantic interpretation.

Final Conclusion

A **CFG + PCFG + Feature Structures + Earley Parsing + Medical Sub-Categorization** architecture is significantly more suitable than basic CFG alone. CFG provides syntactic structure, PCFG resolves ambiguity, feature structures enforce grammatical agreement, Earley parsing improves handling of complex/partial input, and sub-categorization ensures that medical actions are correctly connected to patients, treatments, and other entities.